MILITARY SPECIFICATION

CONTROL, POWER SUPPLY, LASER: C-9135( )/VVG-1: 11738831

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electronic unit known as Control, Power Supply, Laser: C-9135( )/VVG-1 Part No. (P/N) 11738831 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment; Acquisition, Maintenance, and Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control Systems, Procedures for
MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F8565556	Pressure Tester
F10559295	Circuit Card Assembly (Reply Gating)
F10559305	Circuit Card Assembly (Counters)
F10559345	Power Supply (PFN charge)
F11665526	Connector Cover
F11665528	Ballistic Cover
F11737350	Circuit Card Assembly (Readout)
F11738831	Control, Power Supply, Laser: C-9135( )/VVG-1: 11738831

PACKAGING DATA SHEET

11738831	Packaging of Control, Power Supply, Laser: C-9135( )/VVG-1: 11738831

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Fabrication. The Laser Power Supply Control: C-9135( )/VVG-1, herein referred to as the unit, shall be manufactured in accordance with Drawing F11738831 and drawings pertaining thereto.

3.2 General specification. The unit shall meet the following requirements, where applicable, of MIL-F-13926:

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishes

d. Electrical and electronic assemblies

e. Part identification and marking

f. Workmanship.

3.2.1 Soldering. Soldering shall be in accordance with MIL-S-45743 or MIL-S-46844, except that bare copper wire exposure shall be permitted at clipped ends of component leads.

3.3 Performance. Unless otherwise specified, the unit shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F) and 90°F.

3.3.1 Loads, power, signals, and auxiliary equipment. The unit shall perform as specified herein when the loads, power, signals, and auxiliary equipment of table I are applied as specified.

3.3.1.1 Safety interlock. With primary power not applied, removal of either the top cover of the unit or of the pulse forming network (PFN) shall cause S1 or S2, respectively, to activate, and an open circuit (resistance equal to or greater than one megohm) to exist between J3-D and J4-X. When the cover and the PFN are installed, a closed circuit (resistance equal to or less than 0.3 ohm) shall exist between J3-D and J4-X.

3.3.1.2 Primary power. With the logical zero type-A signal (item 3.2 of table I) applied to J1-q, application of the 24 volt direct current (Vdc) primary power (item 2.1 of table I) shall cause dc voltages to appear at terminals listed in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

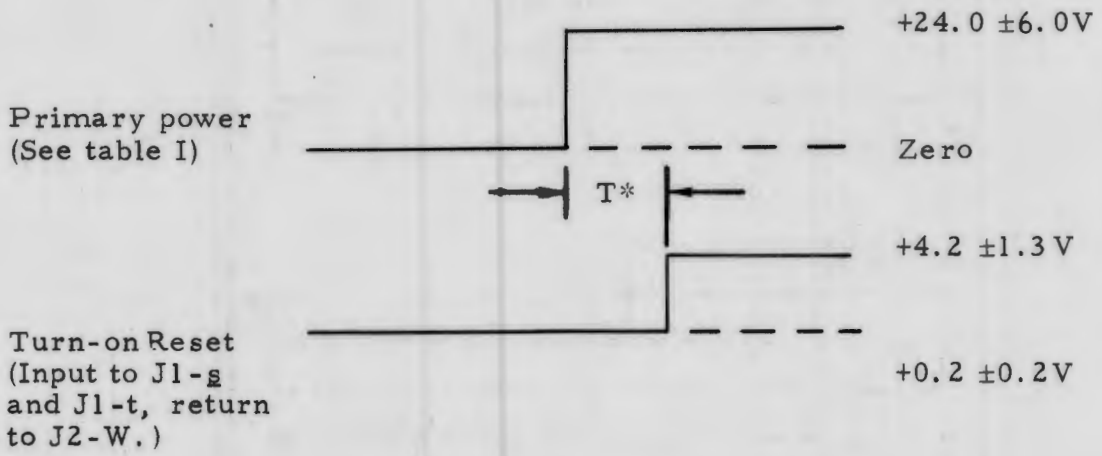
TABLE I
LOADS, POWER, SIGNALS, AND AUXILIARY EQUIPMENT

Item	Conditions	Characteristics	Connections
1.	Loads		Connected between the following pins:
1.1	Resistor	3 megohms ±5 percent, 2500V minimum rating	J2-Z and J2-W
1.2	Resistor	100 kilohms ±1 percent	J2-b and J2-c, unless otherwise specified
1.3	Resistor	32.4 kilohms ±1 percent	J2-b and J2-c, when specified herein
1.4	Resistor	39 ohms ±5 percent	J2-R and J2-W
1.5	Resistor	66.5 kilohms ±1 percent	J5-t and J5-j, when specified herein
1.6	Resistor	300 ohms ±5 percent	J1-X and J2-U when specified herein
2.	Power source		
2.1	+24.0 Vdc (Primary power)	Tolerance: ± 6.0 Vdc Ripple: ≤200 mV p-p	Connected to J4-K and to J4-L, with negative return to J4-M and to J4-N, except when otherwise specified.
3.	Signal sources		
3.1	Remote Range	+8.0 ±0.5 Vdc, in series with 33 kilohms ±5percent	Applied between J5-c(+) and J5-e(-), when specified herein
3.2	Digital type-A	Logical one: +4.0 ±1.0V Logical zero: +0.2 ±0.2V Source impedance: 6kΩ ±15 percent logical one level Pulse width and pulse repetition rate, variable	Applied to input terminals as specified herein
3.3	Digital type-B	Logical one: +4.0 ±1.0V Source impedance: 150Ω ±30 percent Logical zero: +0.2 ±0.2V Pulse width and pulse repetition rate as specified herein	Applied to input terminals as specified herein
3.4	Turn-on Reset	See figure 1	Applied as specified on figure 1, whenever primary power (item 2.1) is applied

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

TABLE I (Continued)

Item	Conditions	Characteristics	Connections
4.	Auxiliary equipment		
4.1	Circuit Card Assembly (Readout)	Drawing F11737350 or equivalent	Connected as shown on figure 2
4.2	Range Data Output Generation Circuit	See figure 3	Connected as shown on figure 3 in lieu of the Circuit Card Assembly (Counters): 10559305, when specified herein



*10 msec <T< 300 msec

Figure 1. Turn-on Reset Waveforms

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

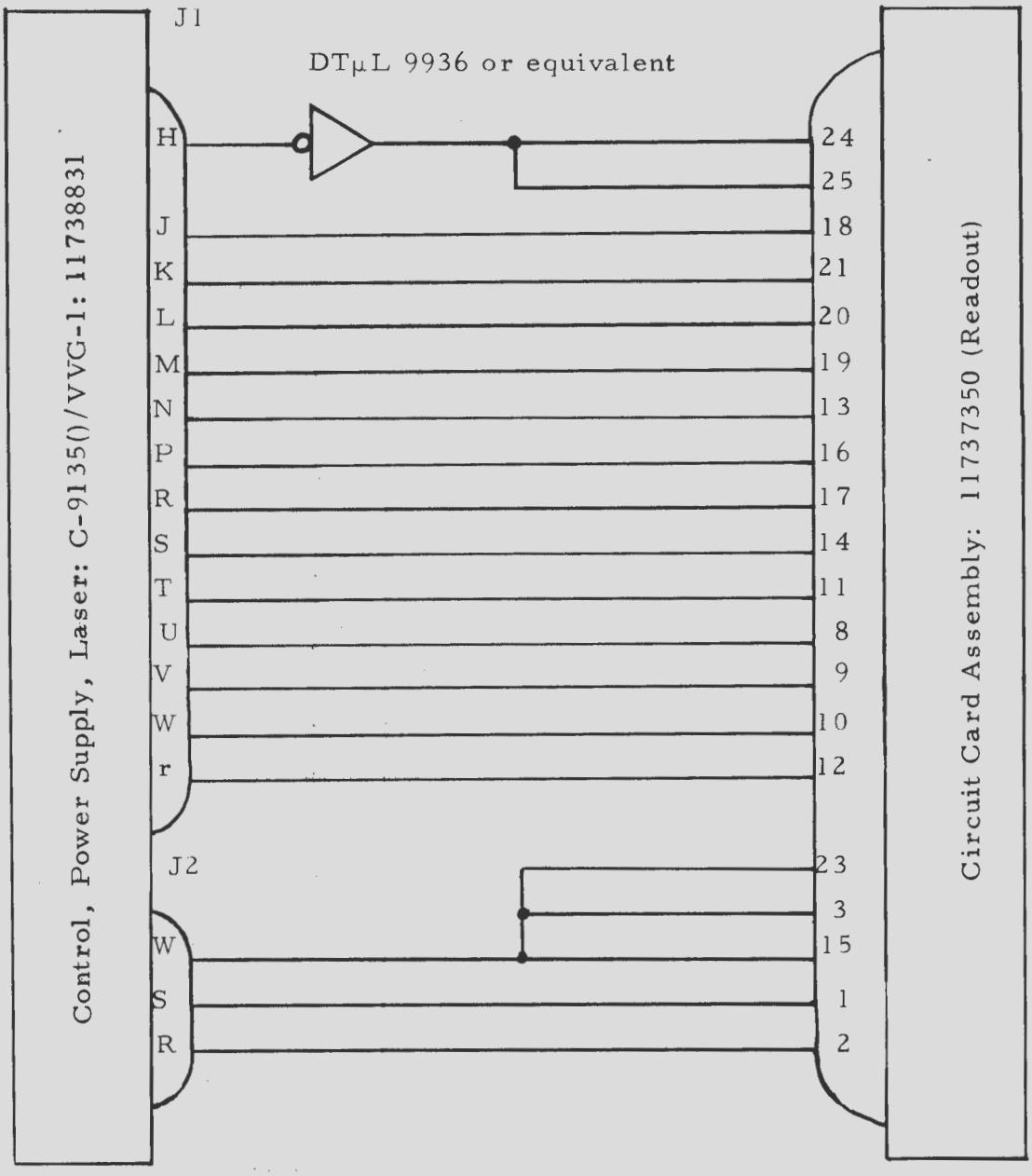
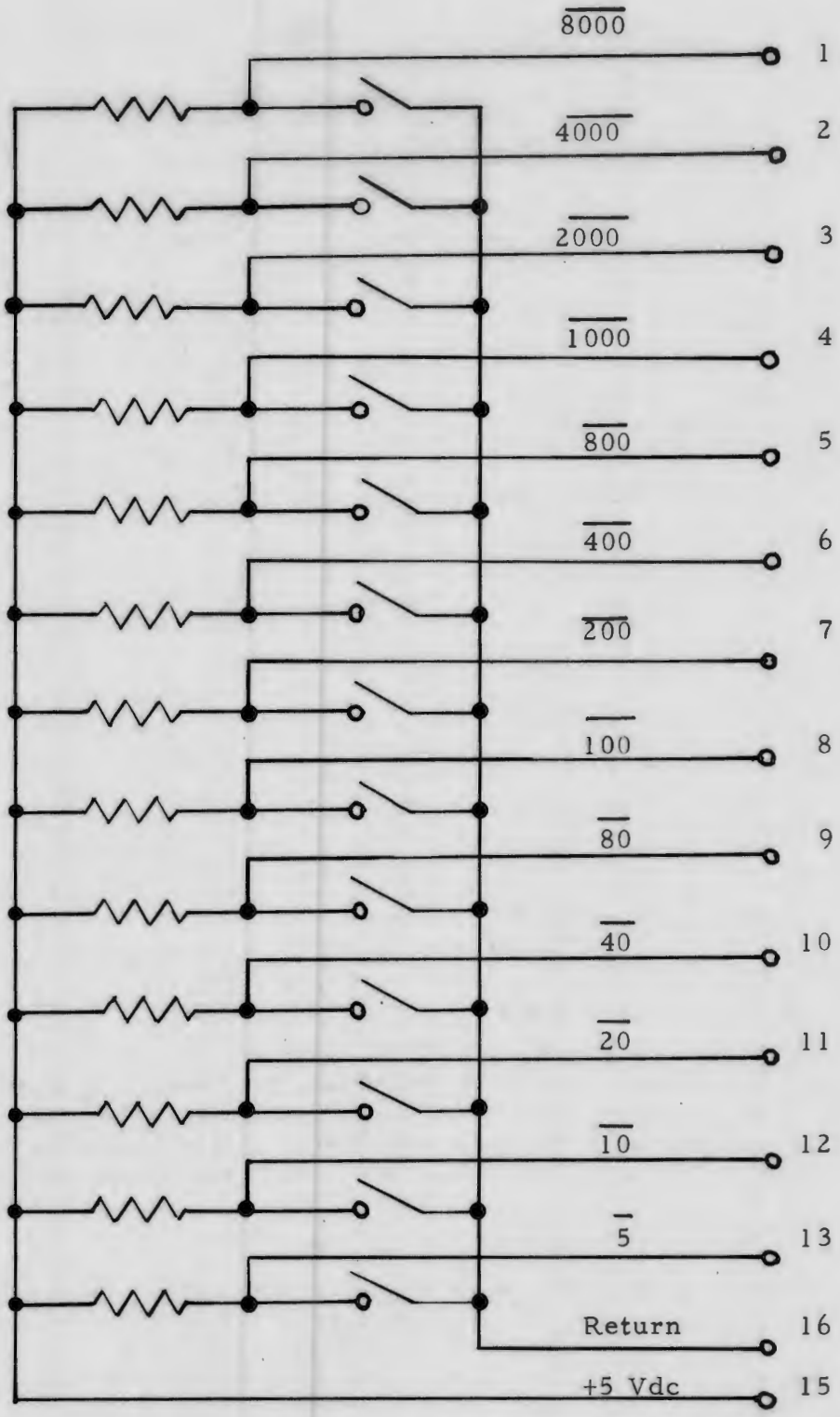


Figure 2. Range Indicator Interconnections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.



NOTES: 1. All resistors are 2 kilohms ±5 percent, 1/4 Watt
2. Pin numbers refer to counter card connector W1A1XA5

Figure 3. Range Data Output Generation Circuit Connection

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

TABLE II
PRIMARY POWER APPLICATION RESPONSE

Item	Output Pins	Output Voltages
1	J2-R	+10.0 ±1.0 Vdc
2	J2-S	+5.0 ±1.0 Vdc
3	J2-T	-6.0 ±1.0 Vdc
4	J2-U	+15.0 ±2.0 Vdc
5	J2-V	+29.0 ±3.0 Vdc
6	J2-Z	-2000 ±300 Vdc
7	J3-A	0 ±10.0 Vdc
8	J4-R	+5.0 ±1.0 Vdc
9	J4-S	-6.0 ±1.0 Vdc
10	J5-i	+10.0 ±1.0 Vdc
11	J5-h	+5.0 ±1.0 Vdc
12	J5-g	-6.0 ±1.0 Vdc
13	J5-j	+15.0 ±2.0 Vdc
14	J5-k	+29.0 ±3.0 Vdc
15	J5-f	+23.0 ±6.0 Vdc

NOTE: Output voltages are referenced to J5-e, J4-M, or J2-W.

3.3.1.3 Range readout. With the range data output generation circuit (item 4.2 of table I) substituted for the Circuit Card Assembly: 10559305 (Counters), closing the switches one at a time, shall cause the range indicated on the signal line identified with the closed switch on figure 3 to be displayed on the Circuit Card Assembly: 11737350 (Readout).

3.3.1.4 Target selection and reset. With J1-j connected to J1-e and J1-f connected to J1-k, and with the input conditions of items 1 through 11 of table III established in the item number order, the J1 outputs and the range readouts shall be as specified in table III.

3.3.1.5 Minimum range inhibit (2). With J1-j connected to J1-e and J1-f connected to J1-k, with the logical on type-B signal of table I applied to J1-h, J1-AA, and J2-h, with the logical zero type-B signal of table I applied to J1-q, J1-g, and J1-z, and with switch S1 (on the Reply Gating Circuit Card Assembly: 10559295) not depressed, the output at J2-1 shall be as shown on figure 4.

Source: https://assist.daml -- Downloaded: 2020-10-22T20:36Z
Check the source to verify that this is the current version before use.

TABLE III
TARGET SELECTION AND RESET RESPONSES

Item	Inputs (1)(2)											Outputs					
	Pins of J1										Pin J2-h	Pin J5	Pins of J1 (1)(3)				Range Readout (±15)
	g	h	m	n	p	s	t	q	z	AA		s	X	Y	Z	a	
1	0	1	0	0	1	1 (4)	1 (4)	0	0	1	1	H	H	L	L	H	0000
2	0	1	0	0	1	1	1	0	0	1	1 (4)	NA	H	L	L	L	830
3	0	1	0	1	0	1	1	0	0	1	1	NA	H	L	L	L	9995 (5)
4	0	1	0	0	1	1	1	0	0	1	1 (4)	NA	L	H	L	L	1830
5	0	1	0	0	1	1	1	0	0	1	1 (4)	NA	H	H	L	L	2830
6	0	1	0	0	1	1	1	0	0	1	1 (4)	NA	L	L	H	L	2830
7	0	1	1	0	0	1	1	0	0	1	1	NA	NA	NA	NA	NA	830
8	0	1	0	1	0	1	1	0	0	1	1	NA	NA	NA	NA	NA	1830
9	0	1	0	0	1	1 (4)	1 (4)	0	0	1	1	NA	H	L	L	H	0000
10	0	1	0	0	1	1	1	0	0	1	1 (4)	NA	NA	NA	NA	NA	830
11	0	1	0	0	1	1	1	0	1	0	1	L	NA	NA	NA	NA	000

NOTES:
1. All signal voltages are referenced to J2-W.
2. Input signal legend: 1 = logical one, type-A signal
   0 = logical zero, type-A signal.
3. Output condition legend: H = +4.0 ±1.5 Vdc
   L = +0.2 ±0.2 Vdc
   NA = Not applicable.
4. Logical zero momentarily applied.
5. Establishment of items 1 through 3 conditions to be repeated if range readout is 000 or 9990.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

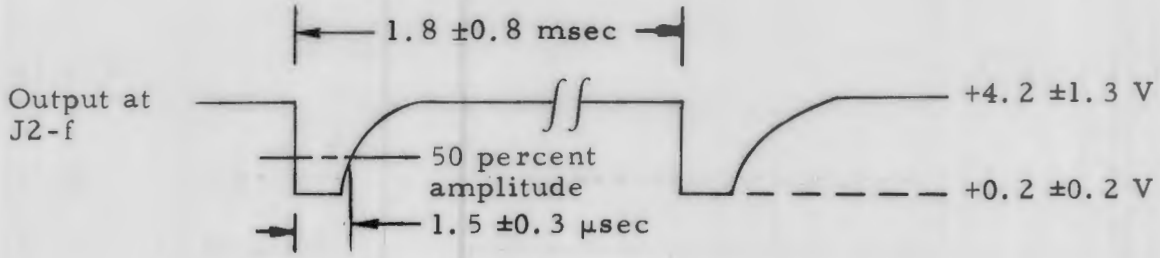


Figure 4. MIN RANGE INHIBIT (2), Waveform

3.3.1.6 A-trigger and video. With the digital type-B signals (item 3.3 of table I) applied as follows:

a. Logical zero to J1-q, J1-n, and J1-p
b. Logical one to J1-AA, J1-z, J1-m, and J2-h
c. Momentary logical zero to J1-t, J1-AA, and J1-s
d. Logical zero to J1-z
e. Voltage of J1-r to read +4.2 ±1.3 V
f. Waveforms of figure 5 to the specified pins,

the range readout shall be 9995. When the range readout indicates 9995, the voltage at J1-r shall switch to +0.2 ±0.2V. If the range indication is 000 or 9990, the waveforms of figure 5 shall be reapplied.

3.3.1.7 Test access output. With the type-B signal (item 3.3 of table I) applied to the input pins as listed in table IV, the corresponding outputs shall be as listed in table IV.

3.3.1.8 PFN charge. With logical one type-B signals applied to J1-q, J2-h and J2-d, with a logical zero type-B signal momentarily applied to J1-t after application of primary power, and with application of primary power (item 2.1 of table I) the signal waveform of figure 6 shall appear at J1-w within 4 seconds of application of power and the waveform at J1-X shall be as shown in figure 6 when the load (item 1.6 of table I) is connected as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

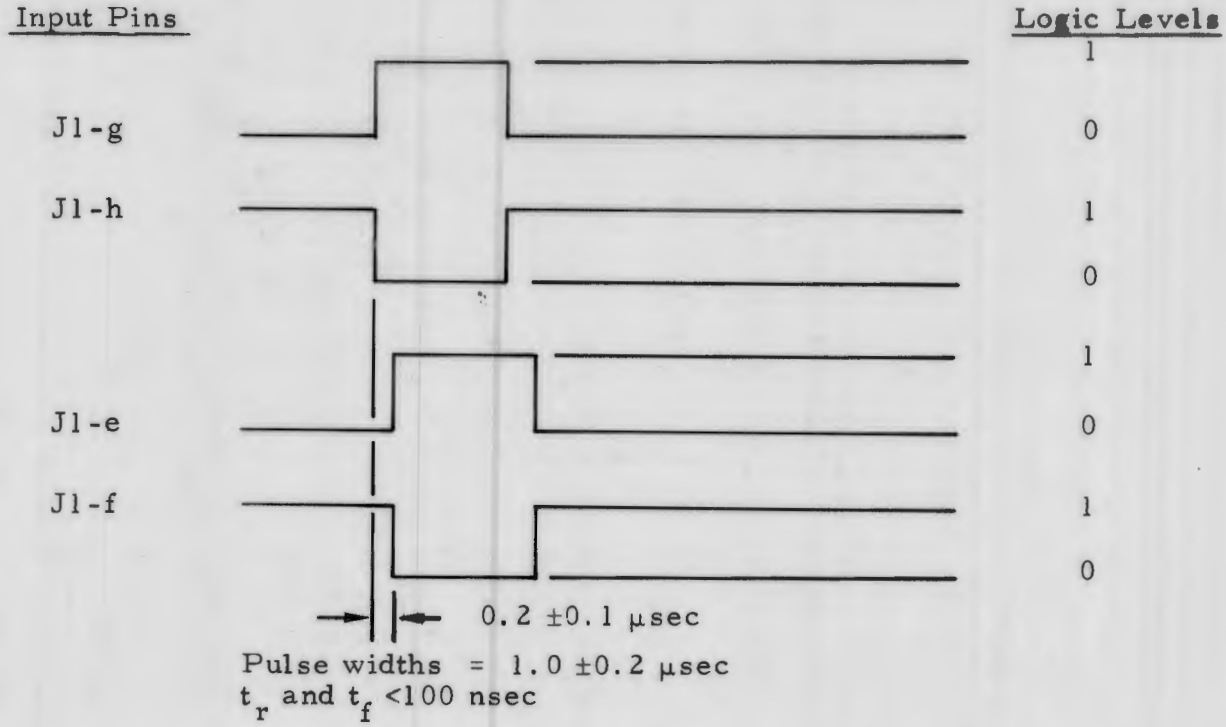


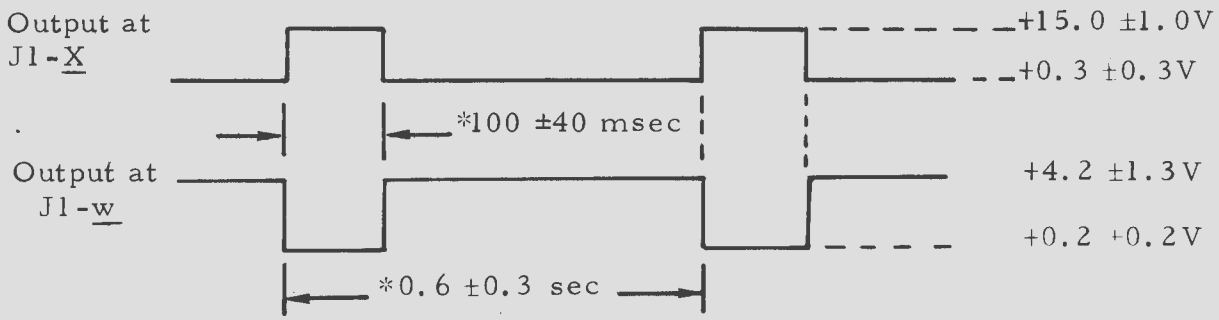
Figure 5. Type-B Signal Input Waveforms -- A-Trigger and Video

TABLE IV
TEST ACCESS OUTPUTS

Item	Input Pins of J1					Output Pins of J5	
	g	h	e	f	q	m	n
1	1	0	NA	NA	0	+0.2 ±0.2 Vdc	NA
2	0	1	NA	NA	0	+4.2 ±1.3 Vdc	NA
3	NA	NA	1	0	0	NA	+0.2 ±0.2 Vdc
4	NA	NA	0	1	0	NA	+4.2 ±1.3 Vdc

NOTE: Symbols are defined in table III

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.



*At the temperature extremes of 3.3.2.2 the pulse width shall be 100 ±60 msec, and the pulse spacing shall be 0.6 ±0.5 sec.

Figure 6. RANGE LIGHT Signal Waveform

3.3.1.9 PFN voltage programming. With the PFN voltage at J3-A set to +1200 ±15V, changing the load between J2-b and J2-c from 100 kilohms (item 1.2, table I) to 32.4 kilohms (item 1.3, table I) shall cause the PFN voltage at J3-A to increase to +1345 ±40V.

3.3.1.10 PFN voltage sensing. With the PFN voltage at J3-A set to +1200 ±15V, the current through a microammeter connected between J5-t and J5-e shall be 75 ±3 microamperes.

3.3.1.11 FIRE signal. With the PFN voltage at J3-A set to +1200 ±200V, with the input signals (items 3.1 and 3.3 of table I) and input pin connection conditions established for the input pins as listed in table V for each item listed therein, and with the table V items conditions established in the order item 1 through item 6, the corresponding ordered outputs at J2-e shall be as listed therein.

3.3.1.12 Malfunction 4 signal.

a. With the PFN voltage at J3-A set to +1200 ±200V, with the input signal (item 3.3 of table I) conditions established for the input pins as listed in table VI for each item listed therein, and with the table VI items conditions established in the order item 1 through item 6, the corresponding ordered outputs at J1-d shall be as listed therein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

TABLE V

$\overline{\text{FIRE}}$ SIGNAL OUTPUTS

Item	Input Conditions at Pins (1)					Outputs at Pin J2-e
	J2-h	J2-d	J1-t	J5-c	J1-q	
1	1	1	1 (2)	B	1	+4.2 ±1.3 Vdc
2	1	1	1	A	1	+0.2 ±0.2 Vdc
3	1	1 (2)	1	B	1	+4.2 ±1.3 Vdc
4	1	1	1 (2)	B	1	+4.2 ±1.3 Vdc
5	1 (2)	1	1	B	1	+0.2 ±0.2 Vdc
6	1	1	1	B	0	+4.2 ±1.3 Vdc

NOTES: 1. Signal/condition legend: 1 = Logical one, type-B signal
0 = Logical zero, type-B signal
A = Remote range signal as specified in table 1
B = Open circuit.
2. Logical zero momentarily applied.

TABLE VI

MALFUNCTION 4 SIGNAL OUTPUTS

Item	Inputs to Pins (1)				Outputs at Pin J1-d
	J1-q	J1-t	J2-d	J2-h	
1	1	1 (2)	1	1	+4.2 ±1.3 Vdc
2	1	1	1	1 (2)	+4.2 ±1.3 Vdc
3	1	1 (2)	1	1	+4.2 ±1.3 Vdc
4	1	1	1 (2)	1	+4.2 ±1.3 Vdc
5	1	1	1	1 (2)	+0.2 ±0.2 Vdc
6	1	1 (2)	1	1	+4.2 ±1.3 Vdc

NOTES: 1. Input signal legend: 1 = logical one, type-B signal.
2. Logical zero momentarily applied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

b. With the 66.5-kilohm resistor (item 1.5 of table I) connected between J5-t and J5-j, the output at J1-d shall be +0.2 ±0.2 Vdc.

3.3.1.13 Malfunction 2 signal. With the digital type-B signals (item 3.3 of table I) applied as follows:

a. Logical one to J1-f, J1-h, and J1-AA

b. Logical zero to J1-e, J1-g, and J1-z

c. Logical zero to J1-q,

momentary application of logical zero to J1-t shall cause the output at J1-c to be +4.2 ±1.3 Vdc. Also, under the signal input conditions of (a), (b), and (c), application of a logical one and a logical zero, simultaneously, to J1-z and J1-AA, respectively, shall cause the output at J1-c to be +0.2 ±0.2 Vdc.

3.3.1.14 Continuity. With no signals or power applied, direct continuity (resistance equal to or less than 0.3 ohm) shall exist between pins listed in table VII.

TABLE VII
CONTINUITY

Item	Pins
1	J1-b and J4-U
2	J1-t and J5-d
3	J1-u and J4-V
4	J1-u and J5-E
5	J1-v and J2-k
6	J1-BB and J5-a
7	J1-CC and J4-Y
8	J2-J and J4-E
9	J2-K and J4-F
10	J2-L and J4-G
11	J2-M and J4-H
12	J2-N and J4-W
13	J2-P and J4-P
14	J2-i and J5-r
15	J2-j and J5-q
16	J2-n and J4-a
17	J4-Z and J5-b

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

3.3.1.15 +400 volt supply. With the PFN voltage at J3-A set to +1200 ±200 V, the voltage at J2-p shall be +400 ±40 Vdc.

3.3.1.16 PFN discharge time. With the PFN voltage at J3-A set to +1200 ± 100 Vdc, and with the subsequent removal of the +24.0 Vdc power source (item 2.1 of table I), the PFN current between J5-t(+) and J5-e(-) shall decrease to less than 7 microamperes within 25 seconds.

3.3.2 Environmental.

3.3.2.1 Storage temperature. The unit shall meet the requirements of 3.3.1 at ambient temperature after exposure to and thermal stabilization at -80° F and +160° F.

3.3.2.2 Operating temperatures. The unit shall meet the requirements of 3.3.1.2 through 3.3.1.16, except 3.3.1.5, while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which the requirements of 3.3.1.2 through 3.3.1.16, except 3.3.1.5, shall be met at ambient temperature.

3.3.2.3 Vibration. The unit shall show no physical failure and shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

a.	Amplitude:	1/64 inch, constant (1/32 inch double amplitude)
b.	Vibratory motion:	Simple harmonic
c.	Sweep time:	5 to 55 to 5 Hertz in a five-minute period
d.	Vibration cycling:	Three linear or logarithmic sweeps applied to each of the threee mutually perpendicular axes.

3.3.2.4 Shock. The unit shall meet the requirements of 3.3.1 after being exposed to a total of 18 shock impulses, three in each direction of the three mutually perpendicular axes. Each shock impulse shall be a half-sine wave with a time duration of 3.5 ±0.5 milliseconds and a peak amplitude of 100g.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

3.3.3 Sealing. With the unit initially pressurized with nitrogen or air to 5 ±1 pounds per square inch gage (psig), the loss of pressure shall not exceed 0.055 psig in one-half hour, except for special sampling environmental during which the loss of pressure shall not exceed 0.22 psig in 2 hours.

3.3.4 Internal pressure. Subsequent to meeting the sealing requirements, the unit shall be dry flushed with an internal atmosphere containing air or nitrogen having a dew point of at least -63.5°F and an internal pressure of 3.5 ±0.5 psig shall be established.

3.3.5 Interface. The unit shall be capable of interfacing with the Ballistic Cover and with the Connector Cover, F11665528 and F11665526, respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three units. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure conformance to all the requirements of this specification. A record of this inspection, including certificates of conformance for materials, shall be submitted with the sample.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified herein or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection and sampling procedures herein and submission of product provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the respective specifications and Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 First article testing. First article testing shall be conducted in accordance with tables VIII through XI to determine compliance with 4.2. All examinations and tests shall be witnessed by the Government Quality Assurance Representative. When manufacturing processes, materials, or inspection techniques previously applied are subject to change, or an adverse change occurs in performance capability of the unit, this shall be cause to cease those operations contributing to the change or failure until corrections have been made by the contractor and approved by the Government.

4.3.2.3 Functional tests. The requirements and tests in table VIII shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature.

4.4 Special sampling. The selection of samples shall be in accordance with section 7 of MIL-STD-105.

4.4.1 General. One unit shall be selected at random by a Government representative as a control sample from each 50 produced. The control sample shall meet the requirements and tests in table IX.

4.4.2 Environmental. Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

TABLE VIII
FUNCTIONAL TESTS

Classification	Characteristics	Requirements	Test Paragraph
101.	Safety interlock	3.3.1.1	4.6.2
102.	Primary power	3.3.1.2	4.6.3
103.	Range readout	3.3.1.3	4.6.4
104.	Target selection and reset	3.3.1.4	4.6.5
105.	$\overline{\text{Minimum range inhibit (Z)}}$	3.3.1.5	4.6.6
106.	A-trigger and video	3.3.1.6	4.6.7
107.	Test access output	3.3.1.7	4.6.8
108.	PFN charge	3.3.1.8	4.6.9
109.	PFN voltage programming	3.3.1.9	4.6.10
110.	PFN voltage sensing	3.3.1.10	4.6.11
111.	$\overline{\text{FIRE}}$ signal	3.3.1.11	4.6.12
112.	$\overline{\text{Malfunction 4}}$ signal	3.3.1.12	4.6.13
113.	$\overline{\text{Malfunction 2}}$ signal	3.3.1.13	4.6.14
114.	Continuity	3.3.1.14	4.6.15
115.	+400 volt supply	3.3.1.15	4.6.16
116.	PFN discharge time	3.3.1.16	4.6.17
117.	Sealing (1/2 hour)	3.3.3	4.6.19

TABLE IX
SPECIAL SAMPLING - GENERAL

Classification	Control Tests	Requirements	Test Paragraph
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

TABLE X
SPECIAL SAMPLING – ENVIRONMENTAL

Classification	Control Tests	Requirements	Test Paragraph
303.	Storage temperature	3.3.2.1	4.6.18.1
304.	Operating temperatures	3.3.2.2	4.6.18.1
305.	Vibration	3.3.2.3	4.6.18.2
306.	Shock	3.3.2.4	4.6.18.3
307.	Sealing (2 hour)*	3.3.3	4.6.19
308.	Internal pressure	3.3.4	4.6.20
309.	Interface	3.3.5	4.6.21
*To be performed after all environmental testing			

TABLE XI
INSPECTION EQUIPMENT

Item	Nomenclature	Description
1.	DC power supply	Plus 24.0 ±2.0 Vdc at 15A Ripple: ≤200 mV p-p maximum
2.	Signal source	Plus 8.0 ±0.5 Vdc Source impedance: 33 kilohms ±5percent
3.	Signal source	Capable of providing the signals as specified in items 3.2 and 3.3, table I
4.	Ohmmeter	Range: 0 to 10 megohms Accuracy: ±5 percent
5.	DC microammeter	Range: 0 to 100 μA Accuracy: ±1 percent Resistance: ≤ 300 ohms
6.	DC voltmeter	Range: 0 to 2500 Vdc Accuracy: ±1 percent
7.	Range data output generator	As shown on figure 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. The inspection equipment listed in table XI shall be utilized in performing the tests specified herein.

4.5.1 Government-furnished inspection equipment. Where the contract provides for Government furnished inspection equipment, supply and maintenance of inspection equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAPs forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the test methods and procedures specified in 4.6 of this specification and with the component inspection procedures specified in examination and test facilities provisions of MIL-F-13926. Since tolerance of inspection equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of inspection equipment shall be so selected and controlled as to insure that the inspection equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of inspection equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

4.5.2.2.1 Pressure tester. Special test equipment (STE) in accordance with Drawing F8565556, or equivalent, shall be provided to check the sealing of the unit.

4.6 Test methods and procedures

4.6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MIL-F-13926.

4.6.1.1 Initial conditions. Unless otherwise specified, connect the loads and power to the unit as specified in table I.

4.6.2 Safety interlock test. With no power applied, measure the resistance between J3-D and J4-X to determine compliance with 3.3.1.1. Remove the unit top cover and again measure the resistance between J3-D and J4-X to determine compliance with 3.3.1.1. Install the unit top cover, then remove the PFN and again measure the resistance between J3-D and J4-X to determine compliance with 3.3.1.1.

4.6.3 Primary power test. Apply the logical zero type-A signal of table I to J1-q, then apply the primary power as listed in table I. Measure the outputs at the pins listed in table II to determine compliance with 3.3.1.2.

4.6.4 Range readout test. Remove the Circuit Card Assembly: 10559305 (Counters) from the unit and substitute the range data output generation circuit (item 4.2 of table I). Separately close each switch on the range data output generation circuit (figure 3) and observe the range readout on the Circuit Card Assembly: 11737350 (Readout) to determine compliance with 3.3.1.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

4.6.5 Target selection and reset test. Connect J1-j to J1-e and J1-f to J1-k, apply the input conditions of table III items 1 through 11 sequentially, measure the corresponding outputs at the listed output pins of J1 and observe the corresponding readouts on the Circuit Assembly: 11737350 (Readout) to determine compliance with 3.3.1.4.

4.6.6 Minimum range inhibit test. Connect J1-j to J1-e and J1-f to J1-k, apply the logical one type-B signal of table I to J1-h, J1-AA, and J2-h, apply the logical zero type-B signal of table I to J1-q, J1-g, and J1-z, and remove the unit cover to release switch S1 on the Circuit Card Assembly: 10559295 (Reply Gating). Observe the output at J2-f to determine compliance with 3.3.1.5.

4.6.7 A-trigger and video test. Apply the digital type-B signal of table I as follows:

a. Logical zero to J1-q, J1-n and J1-p
b. Logical one to J1-AA, J1-z, J1-m, and J2-h
c. Momentary logical zero to J1-t, J1-AA, and J1-s
d. Logical zero to J1-z
e. Voltage of J1-r to read +4.2 ±1.3 V
f. The waveforms of figure 5 as specified.

Observe the readout on the Circuit Card Assembly: 11737350 (Readout) to determine compliance with 3.3.1.6.

4.6.8 Test access output test. Apply the digital type-B signals of table I to the input pins as specified for the items of table IV, and measure the outputs at J5-m or J5-n, as applicable, to determine compliance with 3.3.1.7.

4.6.9 PFN change test. Apply the load (item 1.6 of table I), logical one digital type-B signal of table I to J1-q, J2-h, and J2-d, and primary power (item 2.1 of table I). Momentarily apply a logical zero type-B signal to J1-t after application of primary power. Observe the output signals at J1-w and J1-x to determine compliance with 3.3.1.8.

4.6.10 PFN voltage programming test. Adjust the PFN voltage at J3-A to +1200 ±15 V by means of control R10 on the Power Supply: 10559345

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

(PFN Charge). Change the load between J2-b and J2-c from 100 kilohms to 32.4 kilohms and measure the output at J3-A to determine compliance with 3.3.1.9.

4.6.11 PFN voltage sensing test. Adjust the PFN voltage at J3-A to +1200 ±15 V, connect the microammeter listed in table XI between J5-t and J5-e, and measure the current to determine compliance with 3.3.1.10.

4.6.12 $\overline{\text{FIRE}}$ signal test. Adjust the PFN voltage at J3-A to +1200 ±200 V, and establish the input conditions for the table V items in the items number order. Measure the corresponding ordered outputs at J2-e to determine compliance with 3.3.1.11.

4.6.13 $\overline{\text{Malfunction 4}}$ signal test.

a. Adjust the PFN voltage at J3-A to +1200 ±200 V, and establish the input signal conditions for the table VI items in the items number order. Measure the corresponding ordered outputs at J1-d to determine compliance with 3.3.1.12a.

b. Connect the 66.5 kilohm resistor (item 1.5) of table I as specified, and measure the output at J1-d to determine compliance with 3.3.1.12b.

4.6.14 $\overline{\text{Malfunction 2}}$ signal test. Apply the digital type-B signals (item 3.2) of table I, as follows:

a. Logical one to J1-f, J1-h, and J1-AA

b. Logical zero to J1-e, J1-g, and J1-z

c. Logical zero to J1-q.

Then apply a momentary logical zero to J1-t, and measure the output at J1-c to determine compliance with 3.3.1.13. Repeat application of signals as specified in (a), (b), and (c), and then apply a logical one and a logical zero simultaneously to J1-z and J1-AA, respectively, and measure the output at J1-c to determine compliance with 3.3.1.13.

4.6.15 Continuity test. Remove the primary power and perform resistance measurements between terminal pins as listed in table VII, to determine compliance with 3.3.1.14.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

4.6.16 +400 volt supply test. Adjust the PFN voltage at J3-A to +1200 ±200 V by means of control R10 on the Power Supply: 10559345 (PFN Charge). Measure the voltages at J2-p to determine compliance with 3.3.1.15.

4.6.17 PFN discharge time test. Connect a microammeter between pins t(+) and e(-) on test access connector J5. Remove the power source (item 2.1 of table I). Measure the PFN current decrease to determine compliance with 3.3.1.16.

4.6.18 Environmental tests.

4.6.18.1 Temperature test. The test apparatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926 except that the unit may be mounted on the inside surface of the test chamber. Perform this test in accordance with Procedure I of MIL-F-13926, with conditions C (-40°F) and E (+125°F), and with the rate of temperature change during cycling not greater than 8°F per minute. At each of the specified operating temperatures, perform the tests of 4.6.3 through 4.6.17, except 4.6.6, to determine compliance with 3.3.2.1 and 3.3.2.2.

4.6.18.2 Vibration test. Perform this test utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Part 1 of Procedure I therein, except that the vibratory conditions of 3.3.2.3 shall apply. Following vibration exposure, examine the unit for structural failures, loose hardware, wire breakage, and abrasion. Perform the tests of 4.6.2 through 4.6.17 to determine compliance with 3.3.2.3.

4.6.18.3 Shock test. Perform this test utilizing apparatus specified in Method 516.1 of MIL-STD-810, in accordance with Procedure I therein and with the conditions specified in 3.3.2.4, to determine compliance with 3.3.2.4.

4.6.19 Sealing test. Perform this test utilizing STE specified in 4.5.2.2.1. Pressurize the unit as specified in 3.3.3. Measure the pressure loss over the applicable time period to determine compliance with 3.3.3.

4.6.20 Internal pressure test. Perform this test utilizing the STE specified in 4.5.2.2.1. Verify that the requirement of 3.3.4 is met.

4.6.21 Interface test. This test shall be conducted using a limit gage to determine compliance with 3.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11738831. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Laser Power Supply Control C-9135( )/VVG-1 is a unit of the Fire Control Rangefinder AN/VVG-1 ( ). The unit performs:

a. Power supply functions

b. Energy storage functions

c. Basic ranging functions.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Applicable packaging data sheet number (see 5.1)

c. Selection of applicable levels of preservation, packaging, and packing

d. Applicable stock number

e. Requirement for First Article Submission.

Custodian:
Army - MU

Preparing activity:
Army - MU

PROJECT NO. 1240-A311

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.